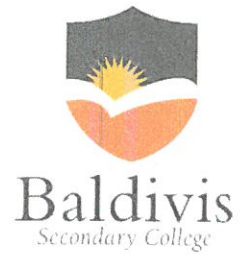


Year 10
Pathway 2



Mini Test 1

Indices and Algebraic fractions

Name: SOLUTIONS

Score: /44
Total Time: 40 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used.

1 Write $10 \times 10 \times 10$ in index notation [1]

$$10^3$$

2 Evaluate 2^5 [1]

$$2 \times 2 \times 2 \times 2 \times 2 = 32$$

3 Evaluate 6^0 [1]

$$= 1$$

4 Simplify $4^5 \times 4$ leaving your answer in index form. [1]

$$4^6$$

5 Simplify $x^7 \div x^3$ [1]

$$x^{7-3} = x^4$$

6 Simplify $10x^5 \div 5x^2$ [1]

$$\frac{10x^5}{5x^2} = 2x^3$$

7 Simplify $(y^5)^6$ [1]

$$y^{30}$$

8 Simplify $(2x^4y)^3$ [2]

$$2^3 x^{12} y^3 \\ 8x^{12}y^3$$

9 Simplify $4m + 7n - 3m + 2n$. [1]

$$m + 9n$$

10 Expand $4(2x + 5)$. [1]

$$8x + 20$$

11 Expand $3n(n - 3m)$ [1]

$$3n^2 - 9mn$$

12 Expand $-2x(x - 3)$ [1]

$$-2x^2 + 6x$$

13 Factorise $5x + 30$ [2]

$$5(x + 6)$$

14 Factorise $16y + 32xy^3$ [2]

$$16y(1 + 2xy^3)$$

15 Expand $(x + 8)(x + 1)$. [2]

$$x^2 + 9x + 8$$

16 Expand $(3t - 2)(4t + 11)$. [2]

$$12t^2 - 8t + 33t - 22$$
$$12t^2 + 25t - 22$$

17 Expand $(a - 5)^2$. [2]

$$(a - 5)(a - 5)$$
$$a^2 - 10a + 25$$

18 Simplify and leave your answer in index form [1]

$$\frac{2^4 \times 3^3 \times 5^8}{5^2 \times 2^3}$$

$$2 \times 3^3 \times 5^6$$

19 Simplify and leave your answer in index form [3]

$$\frac{(3^2 \times 2^6)^4}{2^8 \times 3}$$

$$\frac{3^8 \times 2^{24}}{3 \times 2^8} = 3^7 2^{16}$$

20 Simplify and leave your answer in index form [3]

$$\frac{4x^{-2} \times 4^2}{4^{-1}}$$

$$4^1 x^{-2} \times 4^3$$
$$4^4 x^{-2}$$

21 Simplify and leave your answer in index form [2]

$$9p^2q^5 \div 27p^6q^5$$

$$\frac{9p^2q^5}{27p^6q^5} = \frac{1}{3p^4}$$

$$= \frac{1}{3p^4}$$

22. Simplify and leave your answer in index form [1]

$$\frac{x}{7} + \frac{4x}{7} = \frac{5x}{7}$$

23 Simplify [3]

$$\frac{x-2}{3} + \frac{2x+5}{3}$$

$$\frac{3x+3}{3} = \frac{3(x+1)}{3}$$

$$= x+1$$

24 Simplify [1]

$$\frac{a}{5} \times \frac{4b}{3}$$

$$\frac{4ab}{15}$$

25 The product of a and b is added to the reciprocal of the sum of a and b .

(a) Write down the expression that describes this information [1]

$$ab + \frac{1}{a+b}$$

(b) Write the expression in part (a) with a common denominator. [3]

$$ab + \frac{1}{a+b}$$

$$= \frac{ab(a+b)}{a+b} + \frac{1}{a+b}$$

$$= \frac{a^2b + ab^2 + 1}{a+b}$$

26 Simplify [3]

$$\frac{4m}{7} + \frac{2m}{3}$$

$$\frac{3(4m) + 7(2m)}{21}$$

$$\frac{12m + 14m}{21}$$

$$\frac{26m}{21}$$

End of Test